



9.2. Data Distribution, Skewness, and Kurtosis

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Understanding how data is distributed is an important part of exploratory data analysis. The distribution of a dataset reveals how values are spread, whether they are concentrated around a central value, and whether unusual patterns or outliers exist. By examining data distributions, we can choose appropriate statistical methods, detect anomalies, and better understand the characteristics of a dataset.

In this section, we use **histograms** together with **Pandas**, **NumPy**, and **Matplotlib** to visualize distributions and measure their skewness and kurtosis.

We will use this synthetic dataset for this section:

```
import numpy as np
import pandas as pd

np.random.seed(42)
```

```

n = 1000

df = pd.DataFrame({
    "ID": [i for i in range(1, n + 1)],
    "Age": np.random.randint(18, 22, n),
    "Major": np.random.choice(["AI", "CSC", "IT", "SW"],
n),
    "GPA": np.round(np.random.uniform(2.0, 4.0, n), 2)
})

df.groupby('Major')['GPA'].agg('mean')

```

1. Types of Data Distribution

Different datasets follow different probability distributions. Some of the most common distributions encountered in data analytics include:

- **Uniform Distribution:** Every value within a specified range has an equal probability of occurring.
- **Binomial Distribution:** Models the number of successes in a fixed number of independent trials.
- **Poisson Distribution:** Models the number of events occurring within a fixed interval of time or space.
- **Normal Distribution:** A symmetric, bell-shaped distribution where most observations cluster around the mean.
- **Exponential Distribution:** Models the time between independent events that occur continuously at a constant average rate.
- **Skewed Distribution:** An asymmetric distribution with a longer tail on either the left or right side.
- **Bimodal Distribution:** Contains two distinct peaks, often indicating that the data consists of two different groups.
- **Log-Normal Distribution:** A distribution in which the logarithm of the data follows a normal distribution. It commonly appears in income, biological measurements, and growth-related data.

The following code generates several common distributions using NumPy.

```
import numpy as np
import matplotlib.pyplot as plt

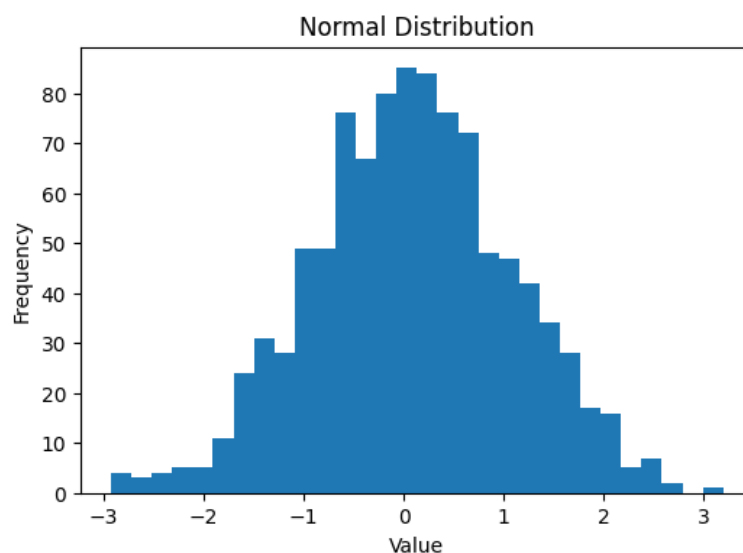
np.random.seed(42)

uniform = np.random.uniform(0, 1, 1000)
normal = np.random.normal(0, 1, 1000)
exponential = np.random.exponential(1, 1000)
binomial = np.random.binomial(10, 0.5, 1000)
poisson = np.random.poisson(5, 1000)
lognormal = np.random.lognormal(0, 0.5, 1000)
```

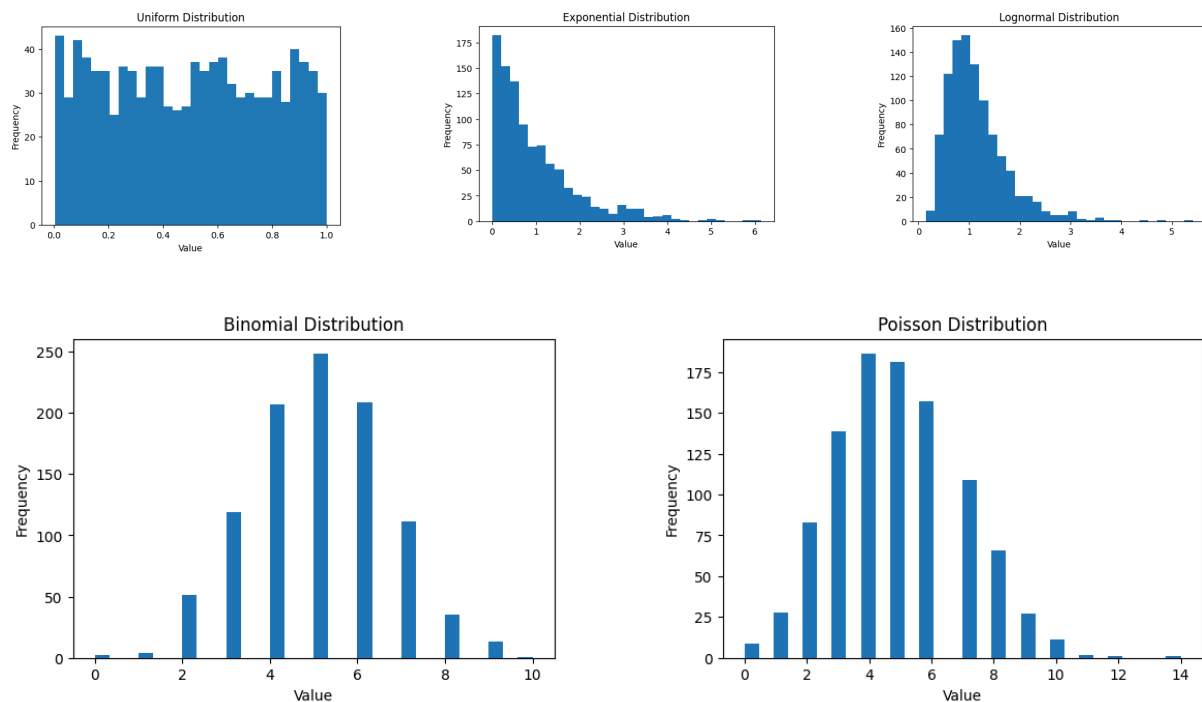
For example, a histogram of a normally distributed dataset can be plotted as follows:

```
import matplotlib.pyplot as plt

plt.figure(figsize=(6,4))
plt.hist(normal, bins=30)
plt.title("Normal Distribution")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()
```

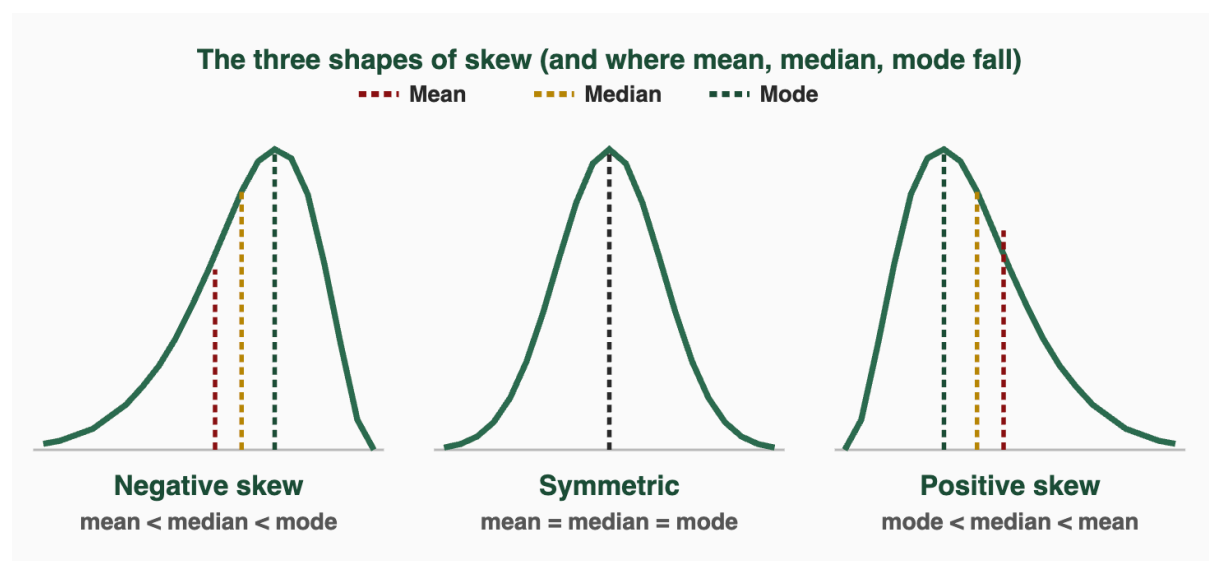


You may replace `normal` with any of the other generated datasets (`uniform` , `exponential` , `binomial` , `poisson` , or `lognormal`) to observe their different shapes.



2. Skewness

Skewness measures the asymmetry of a data distribution.



- **Skewness > 0:** Right-skewed distribution with a longer tail on the right.

- **Skewness < 0:** Left-skewed distribution with a longer tail on the left.
- **Skewness $\approx$ 0:** Approximately symmetric distribution.

Pandas can compute skewness directly.

```
print("GPA Skewness:", df["GPA"].skew())
```

To visualize the distribution, plot a histogram.

```
plt.figure(figsize=(6,4))
plt.hist(df["GPA"], bins=8)
plt.title("Histogram of GPA")
plt.xlabel("GPA")
plt.ylabel("Frequency")
plt.show()
```

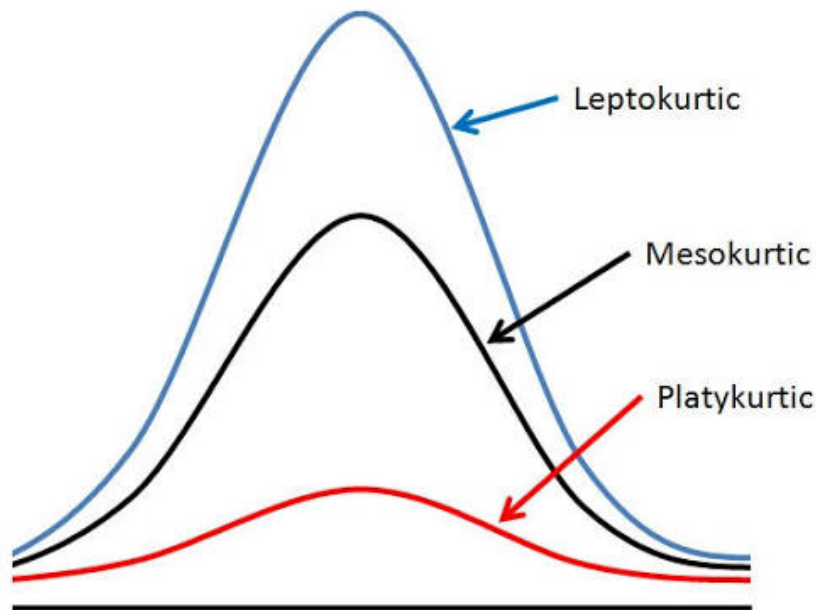
A simple interpretation can be performed automatically.

```
skew = df["GPA"].skew()

if skew > 0.5:
    print("The distribution is positively (right) skewed.")
elif skew < -0.5:
    print("The distribution is negatively (left) skewed.")
else:
    print("The distribution is approximately symmetric.")
```

3. Kurtosis

Kurtosis measures the heaviness of a distribution's tails and the sharpness of its peak.



- **Kurtosis > 3:** Sharp peak with heavier tails (more extreme values).
- **Kurtosis < 3:** Flatter distribution with lighter tails.
- **Kurtosis $\approx$ 3:** Similar to a normal distribution.

Pandas reports **excess kurtosis**, where a normal distribution has a value close to **0**.

```
print("Excess Kurtosis:", df["GPA"].kurt())
```

The following code interprets the result.

```
kurt = df["GPA"].kurt()

if kurt > 0:
    print("The distribution has heavier tails than a normal distribution.")
elif kurt < 0:
    print("The distribution has lighter tails than a normal distribution.")
else:
    print("The distribution is similar to a normal distribution.")
```

Summary

In this section, we explored the concept of data distributions and learned how to analyze them using histograms and statistical measures. Specifically, we learned how to:

- Generate several common probability distributions using NumPy.
- Visualize data distributions using histograms.
- Distinguish between uniform, normal, exponential, skewed, bimodal, and log-normal distributions.
- Measure the asymmetry of a dataset using **skewness**.
- Measure the shape of a distribution using **kurtosis**.
- Interpret skewness and kurtosis values to better understand the characteristics of a dataset.

These techniques provide valuable insights into the structure of data and help determine whether additional preprocessing or transformations are needed before further statistical analysis or machine learning.

References

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<https://www.geeksforgeeks.org/data-science/skewness-measures-and-interpretation/>

<https://www.geeksforgeeks.org/maths/how-to-calculate-kurtosis-in-statistics/>

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Next Section:

 [9.3. Covariance and Correlation in Data](#)